ADITYA DAKSH

AI/ML Engineer | Generative AI & LLM Applications | RAG Systems

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PROFILE SUMMARY

Computer Science (AI) graduate seeking AI/ML, Generative AI and software engineering roles. Built a full-stack RAG application tested on 30+ books, plus NLP and computer vision projects in Python, LangChain, FastAPI and PyTorch. Experience deploying models on Docker and NVIDIA DGX A100 infrastructure.

EDUCATION

ABES Institute of Technology, Ghaziabad, Uttar Pradesh *Nov 2022 - Aug 2026*

B.Tech in Computer Science and Engineering (Artificial Intelligence)

Intermediate (Class XII, CBSE) | Shree Mata Prasad Kalavati Devi Shiksha Sansthan, Firozabad, Uttar Pradesh *2021 - 2022*

High School (Class X, CBSE) | Shree Mata Prasad Kalavati Devi Shiksha Sansthan, Firozabad, Uttar Pradesh *2019 - 2020*

PROJECTS

Nexora – RAG Book Assistant | Python, FastAPI, LangChain, ChromaDB, Groq, React, TypeScript | GitHub *Feb 2026 - Oct 2026*

- Built a full-stack Retrieval-Augmented Generation (RAG) application for conversational querying of PDF books, tested on 30+ books, using semantic search and context-grounded LLM responses with page-level source citations.
- Implemented an ingestion pipeline with PyPDFLoader, OpenAI embeddings, and ChromaDB, using configurable chunking and MMR retrieval for diverse, relevant context; integrated Groq for LLM inference.
- Built an assessment engine generating MCQ and fill-in-the-blank questions with semantic answer validation, difficulty levels, mastery scoring, and feedback, served via FastAPI with a React and TypeScript frontend.

BiasSpectra | Python, BERT, Transformers, Streamlit, NLP | GitHub *Aug 2025 - Nov 2025*

- Built an NLP-based political bias detection system for Indian news articles using BERT and rule-based filtering, with a two-stage pipeline that screens out non-political content before classification, working with unstructured text data.
- Developed and deployed an interactive Streamlit application for real-time bias analysis, testing and evaluating model outputs so end users can classify article bias without any setup.

Adaptive Traffic Signal Control System | Python, YOLOv11, SUMO | GitHub *Jan 2025 - Apr 2025*

- Developed a real-time vehicle detection system using YOLOv11, achieving 92% detection accuracy across 4 vehicle classes.
- Simulated adaptive traffic signal control in SUMO to demonstrate congestion reduction and emergency vehicle prioritization logic.

WORK EXPERIENCE

Machine Learning Intern | Ganga Library INC (Remote) *Dec 2025 - Jan 2026*

- Built machine learning classification and trend-analysis models in Python on a dataset of 1,000+ Nobel Prize profiles, handling structured and unstructured data.
- Created visualizations and analytical reports that surfaced key trends to support data-driven decision making.
- Optimized data processing pipelines and debugged workflow issues to reduce execution time and improve efficiency, delivering within defined timelines.

AI/ML Trainee | Global Infoventures Pvt Ltd, Ghaziabad *Sep 2024 - Feb 2025*

- Engineered deep learning models for PPE-violation detection and multi-person face detection applications.
- Managed Linux systems, containerized applications with Docker, and deployed models on NVIDIA DGX A100 infrastructure.
- Led and collaborated with a 4-member team on end-to-end dataset preparation and structured annotation using Roboflow.

TECHNICAL SKILLS

- **Programming Languages:** Python, SQL, Java, JavaScript
- **Generative AI & Agents:** Generative AI, Agentic AI, AI Agents, Large Language Models (LLMs), Prompt Engineering, OpenAI, LangChain, LangGraph, RAG, Embeddings, Vector Databases
- **Machine Learning & AI:** Machine Learning, Deep Learning, PyTorch, TensorFlow, Scikit-learn, NumPy, Pandas, Matplotlib, Transformers, NLP, Computer Vision
- **Deployment & Infra:** REST APIs, JSON, FastAPI, Docker, NVIDIA DGX A100, Linux
- **Databases:** SQL, MySQL, MongoDB, Redis, ChromaDB, Vector Databases
- **Web Technologies:** HTML, CSS, JavaScript, React.js, TypeScript
- **Tools & Platforms:** Git, GitHub, Postman, VS Code, Jupyter Notebook
- **Core CS:** Data Structures & Algorithms, OOP, DBMS, Computer Networks

CERTIFICATIONS & ACHIEVEMENTS

- Java Programming Certification - Soft Pro
- Machine Learning Training Certification - GI Ventures (NVIDIA Partner)
- TCS CodeVita 2024 - TCS
- Python for Machine Learning and Data Science
- Artificial Intelligence and Deep Learning (Academic Training)